

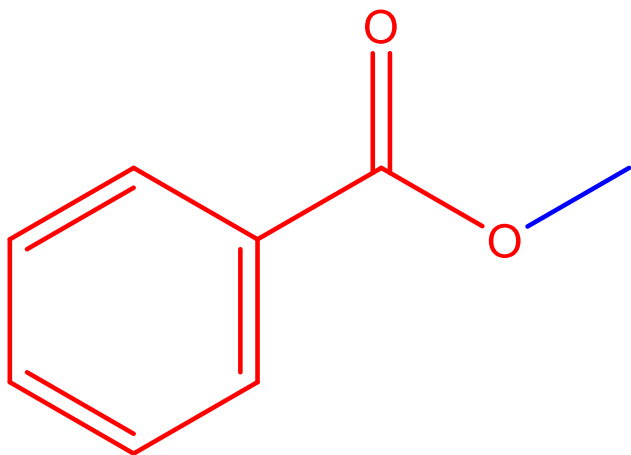
- a) Draw the structure of methyl benzoate and 2-methyl-propyl ethanoate.**
- b) Write the mechanism for the Fischer esterification of benzoic acid with methanol leading to methyl benzoate.**
- c) Discuss why 2-methyl-propyl ethanoate is not synthesised (made) using the same procedure.**

10 Minutes

a) Draw the structure of methyl benzoate and 2-methyl-propyl ethanoate.

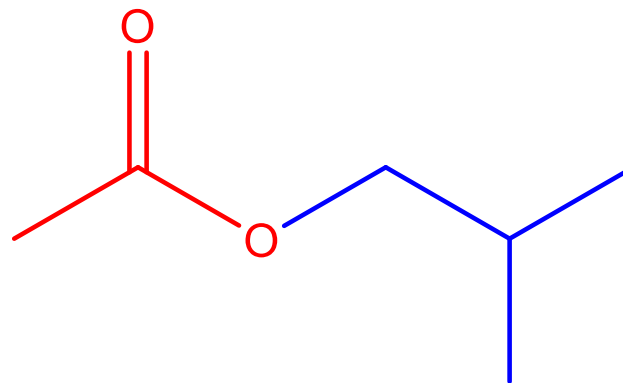
Find the longest chain containing the CO₂- group

Name: alkyl substitute + longest chain containing the CO₂- group with suffix -oate



methyl benzoate

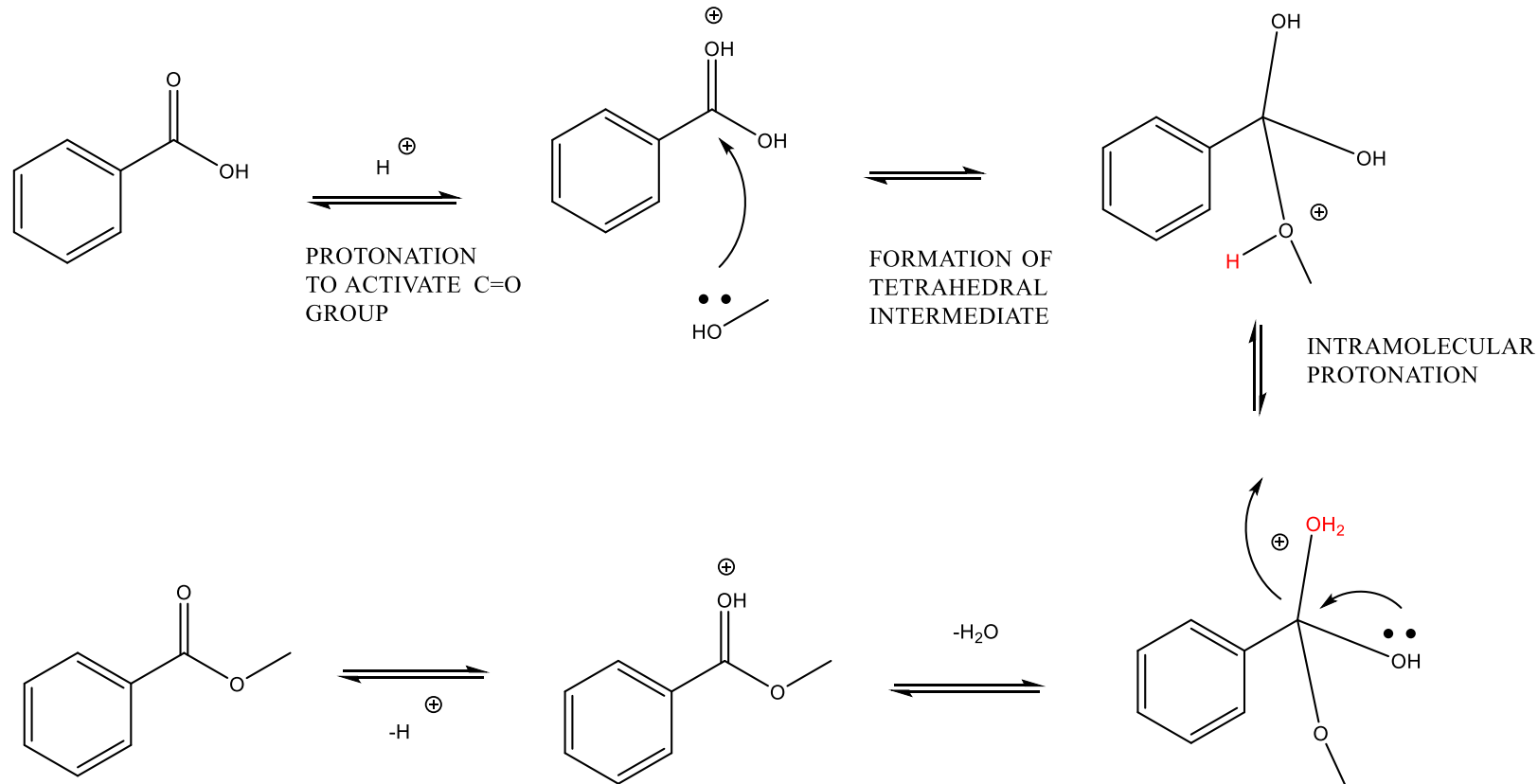
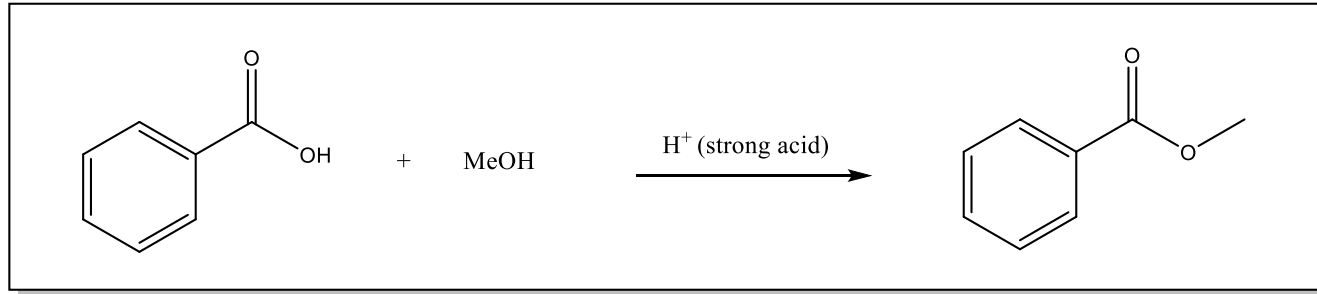
Methyl ester derivative of benzoic acid



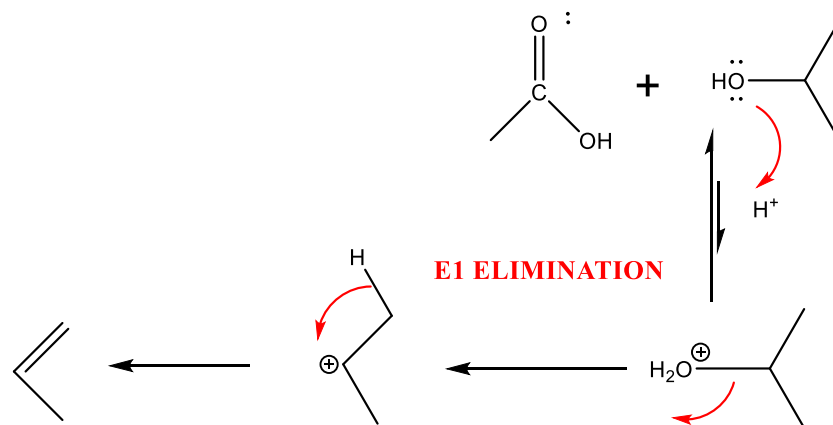
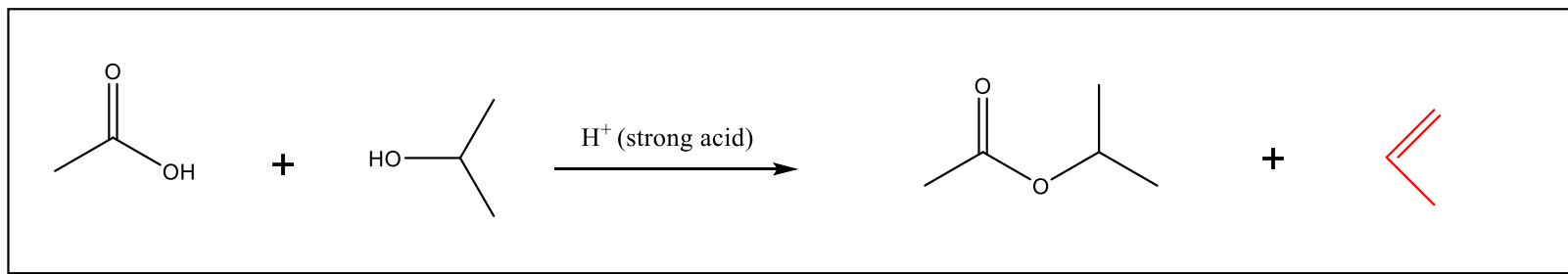
2-methyl-propyl ethanoate

2-Methyl-propyl ester derivative of ethanoic acid

b) Write the mechanism for the Fischer esterification of benzoic acid with methanol leading to methyl benzoate.

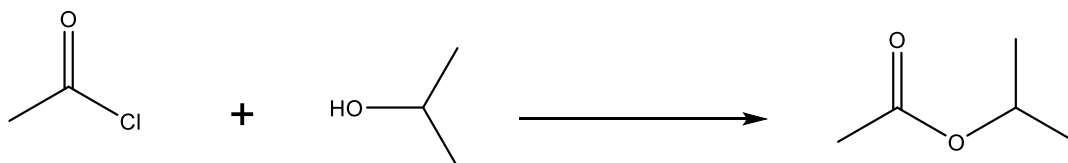


c) Discuss why 2-methyl-propyl ethanoate is not synthesised (made) using the same procedure.

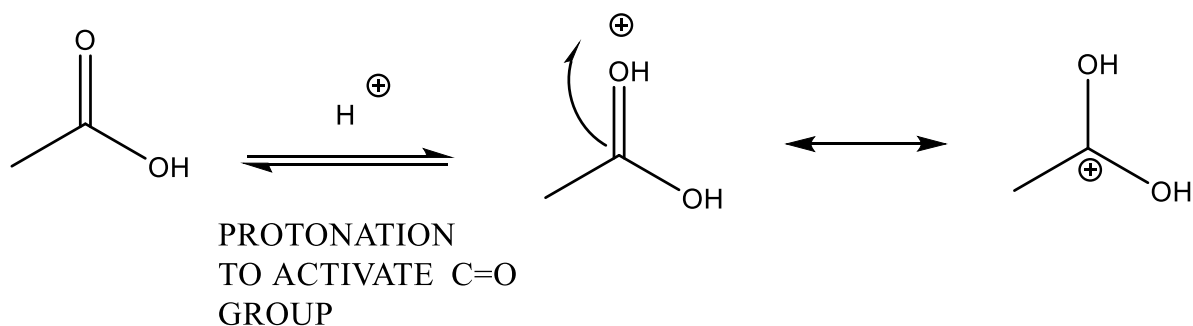
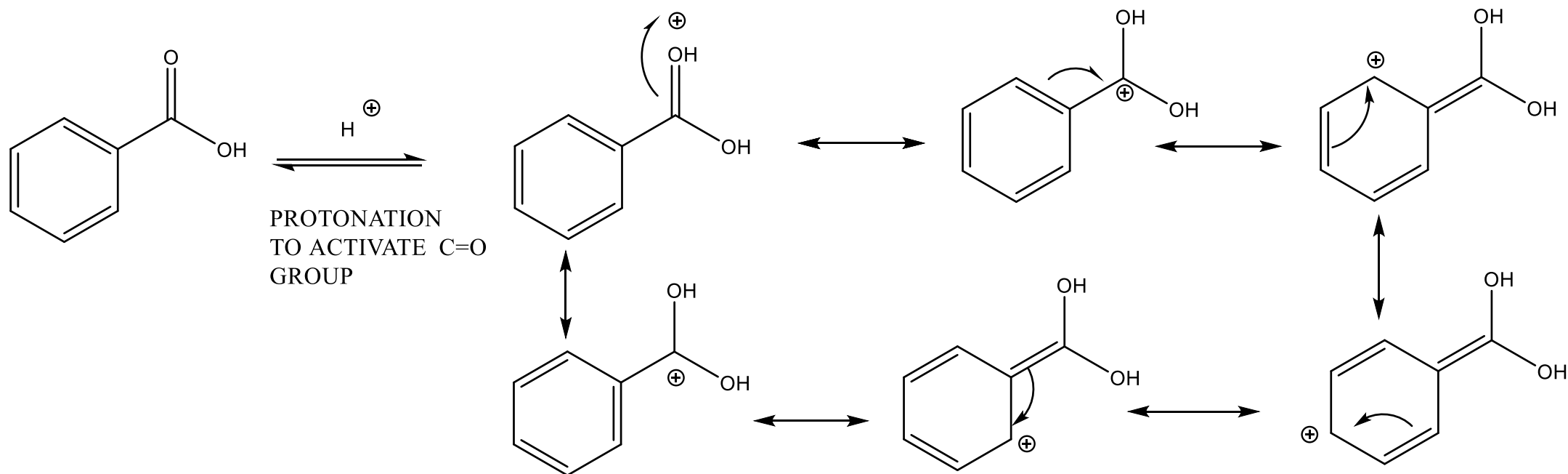


1. 竞争反应
2. 碳正离子不稳定
解决：用相应的酰氯基

THERE IS A SIDE COMPETING REACTION OF E1 ELIMINATION AND FOR THAT REASON IT IS NOT A GOOD ROUTE TO OBTAIN THE PRODUCT. IT WOULD BE BETTER TO START WITH AN ACYL CHLORIDE DERIVATIVE TO AVOID THE PROBLEM



c) Discuss why 2-methyl-propyl ethanoate is not synthesised (made) using the same procedure.

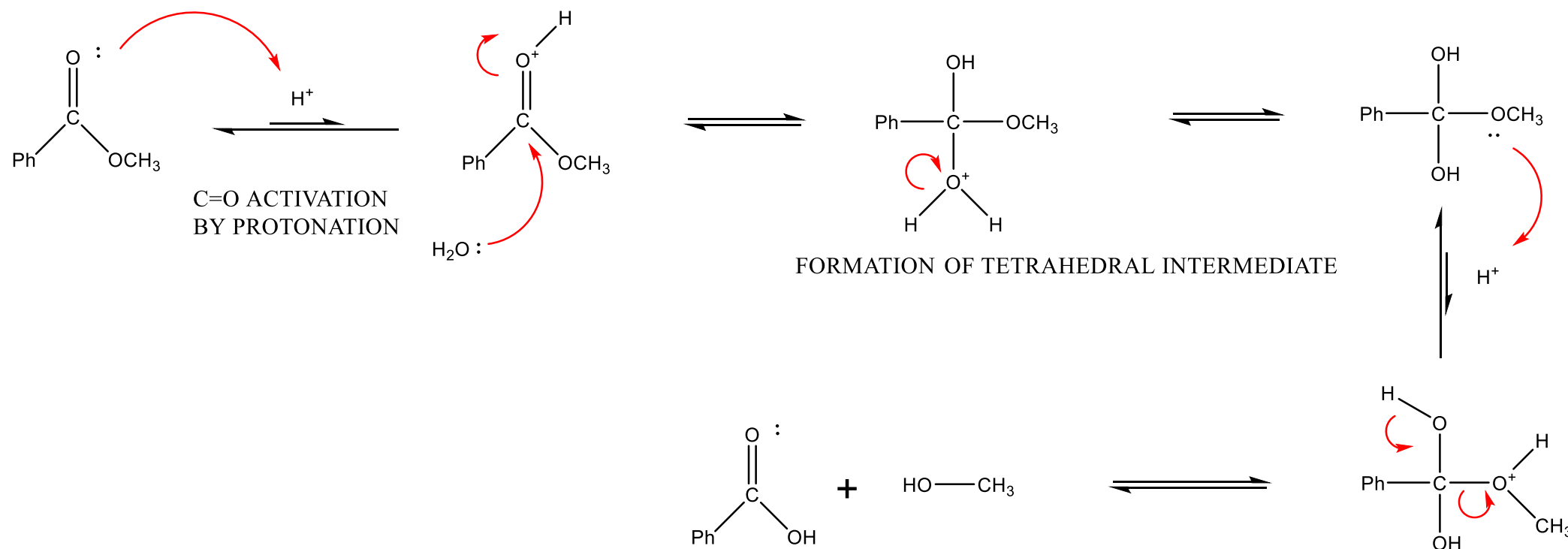
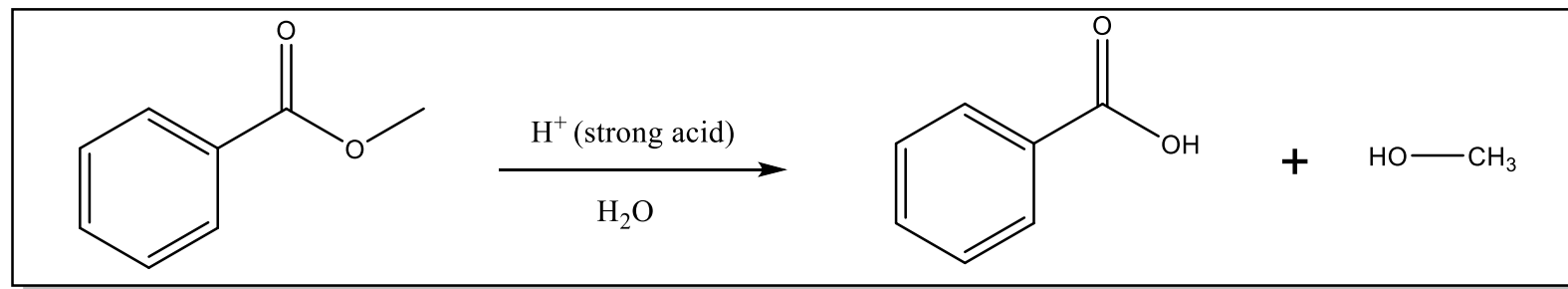
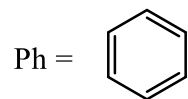


ANOTHER REASON COULD BE THAT THIS PROTONATED INTERMEDIATE IS LESS STABLE, FISCHER SYNTHESIS IS NOT THE BEST ROUTE. IT WOULD BE BETTER TO DO A TRANSESTERIFICATION OR USE THE CORRESPONDING ACYL CHLORIDE DERIVATIVE

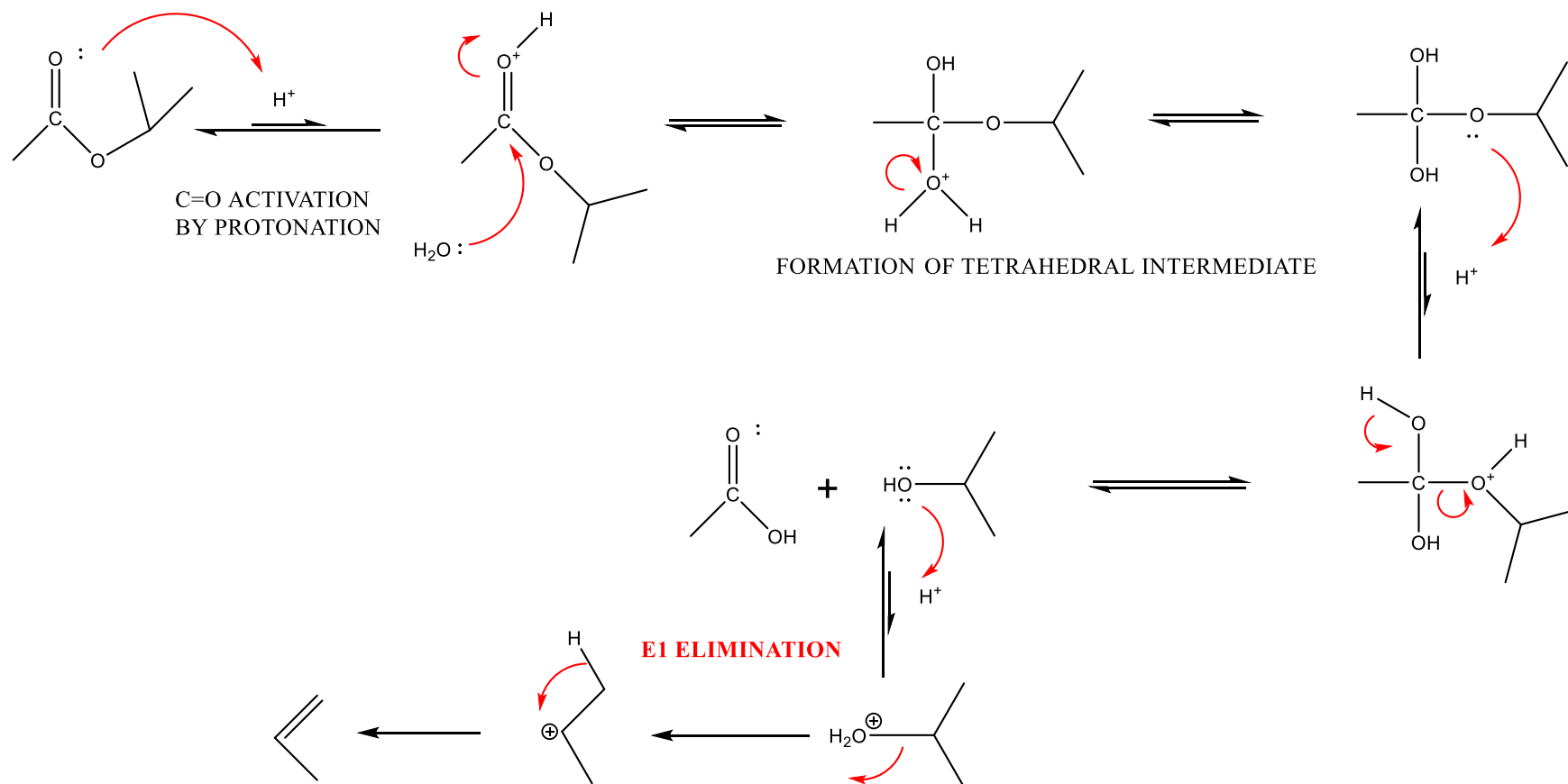
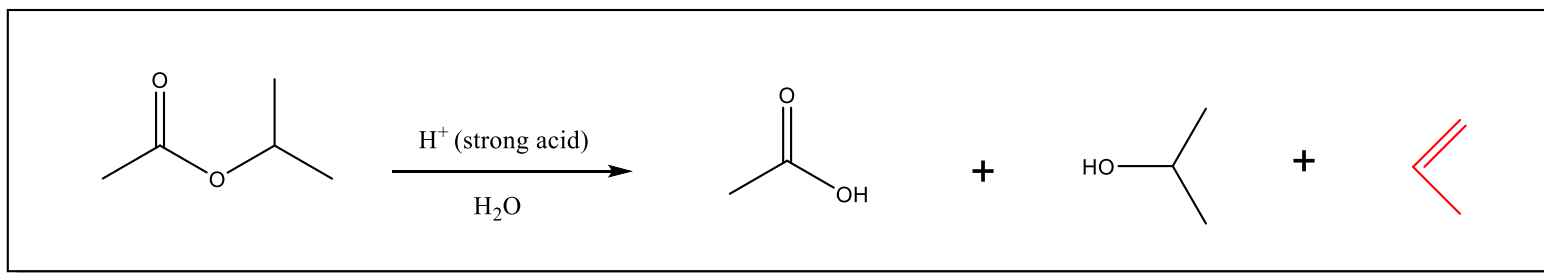
- a) Write the mechanism of the acid hydrolysis of methyl benzoate
- b) When you treat 2-methyl-propyl ethanoate under the same conditions as in 2a, you obtain ethanoic acid(acetic acid) and a mixture of an alkene and 2-methyl-propanol, through a different mechanism. Write this mechanism.

10 Minutes

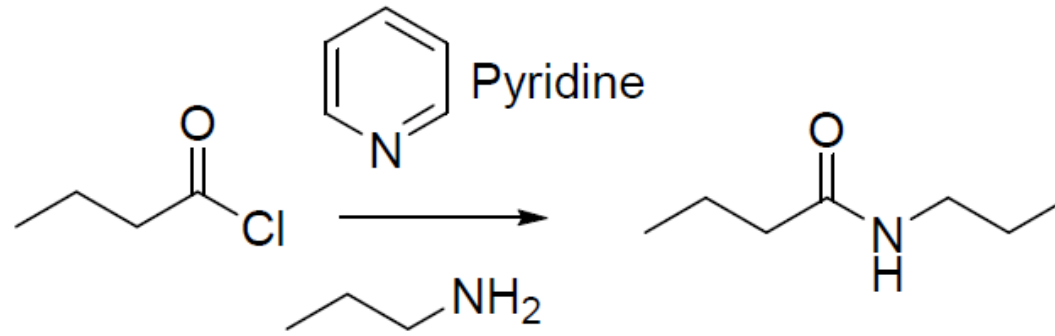
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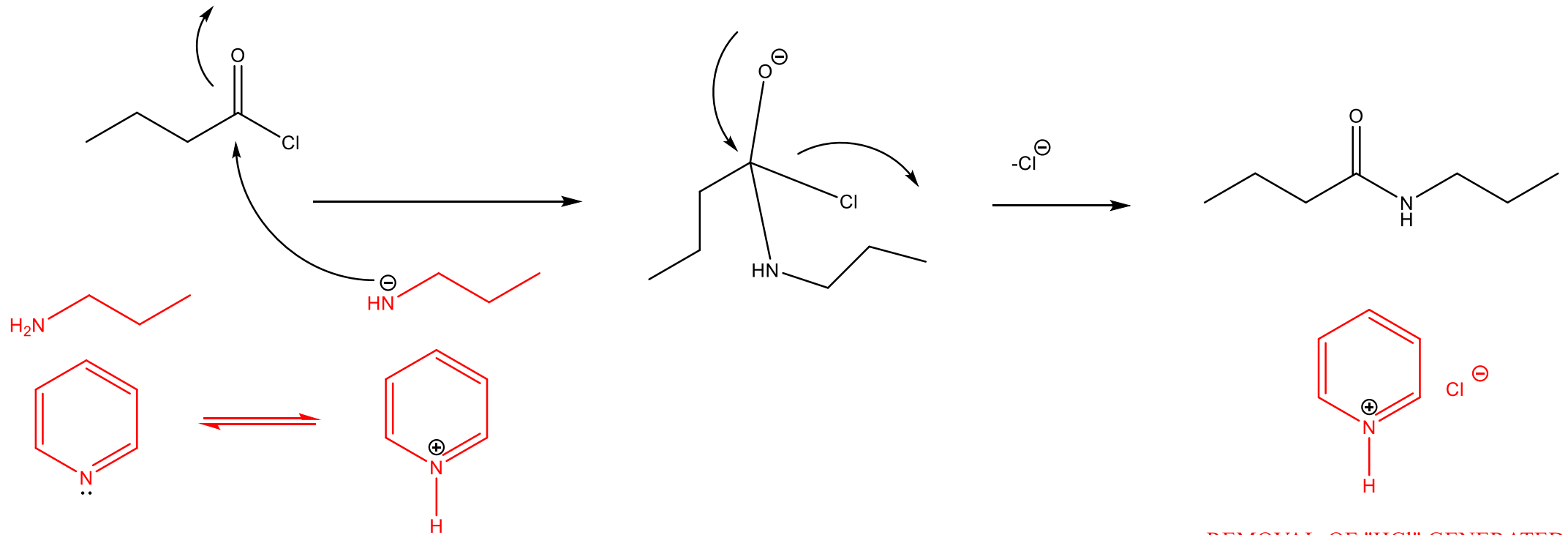
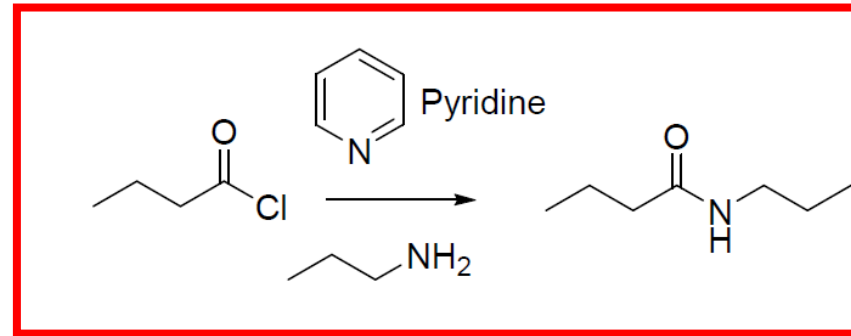
- a) Write the mechanism of the reaction of butanoyl chloride with propanamine, in the presence of pyridine shown below.



- b) Explain why pyridine is used in the system (If pyridine was not used, what would be the impact on the reaction yield and why).

10 Minutes

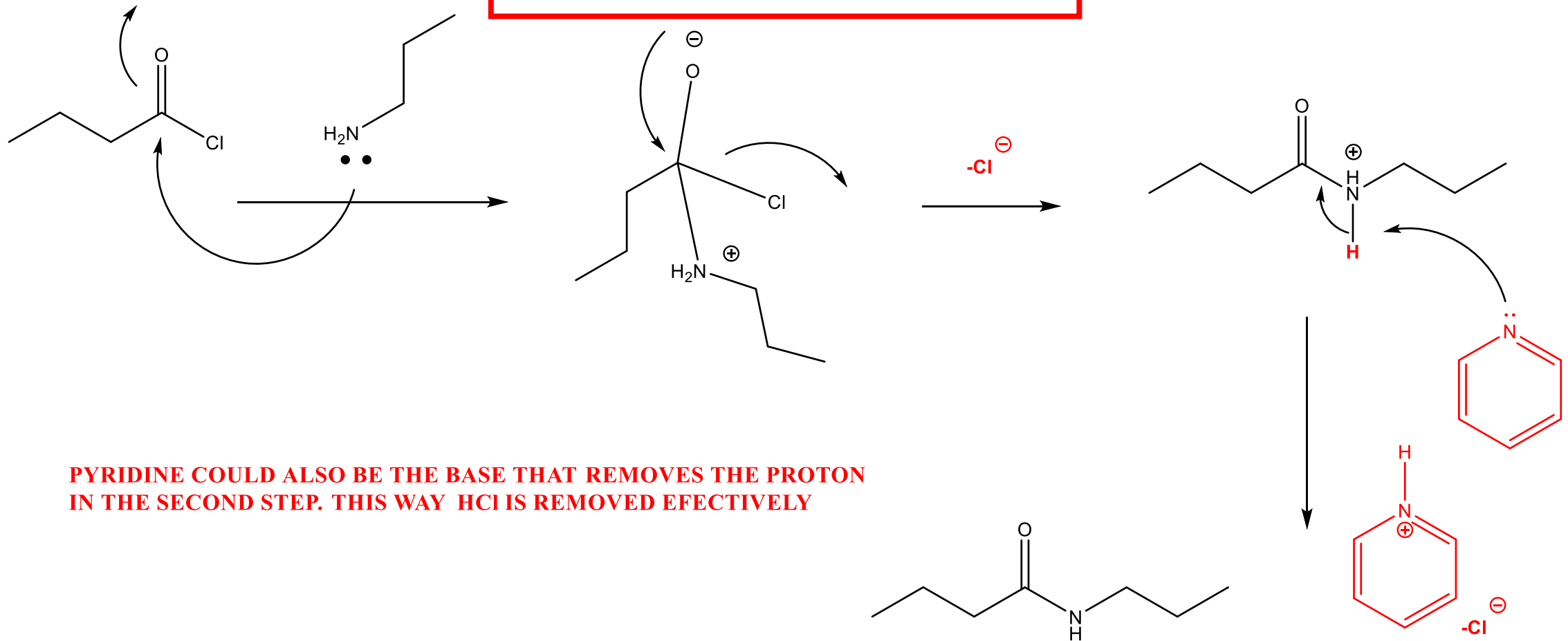
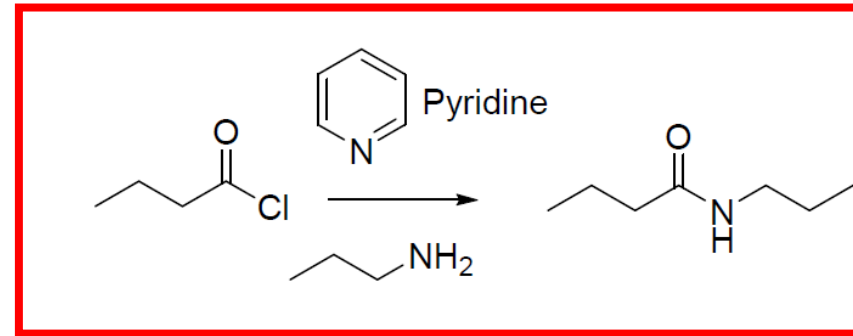
a) Write the mechanism of the reaction of butanoyl chloride with propanamine, in the presence of pyridine shown below.



DEPROTONATED AMINE IS A BETTER NUCLEOPHILE

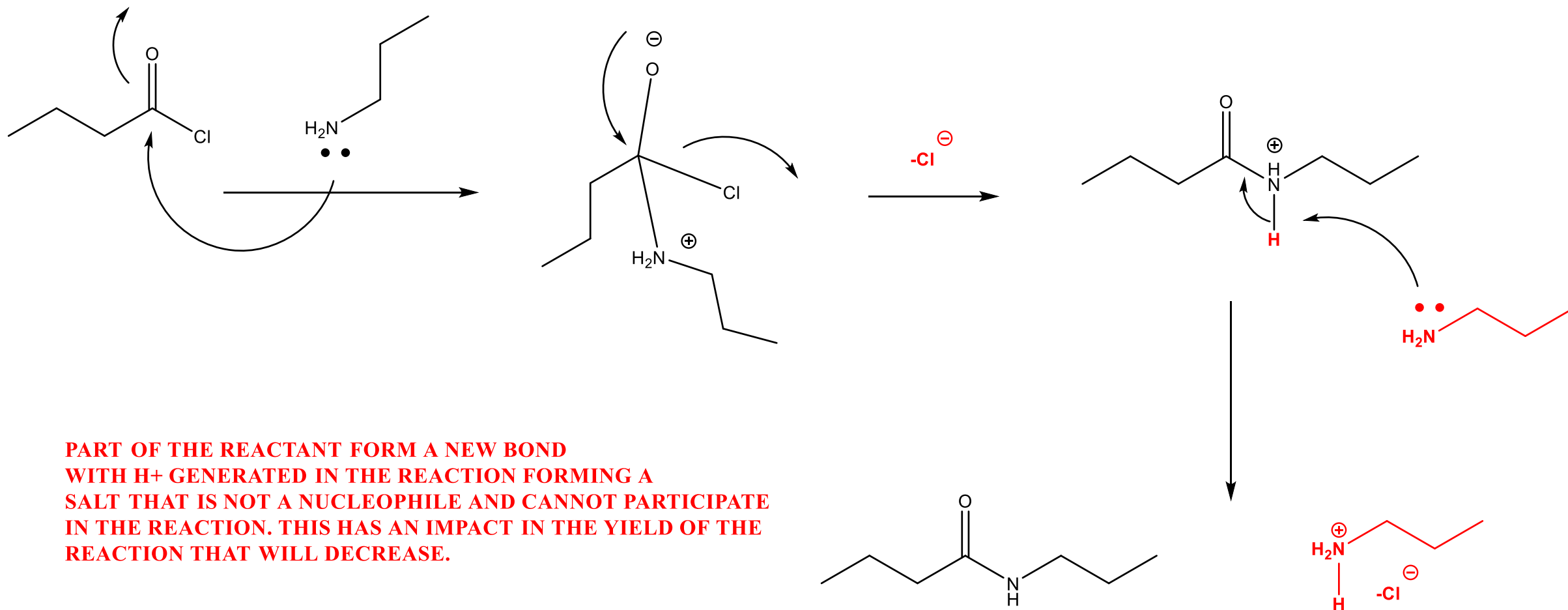
REMOVAL OF "HCl" GENERATED DURING THE REACTION

- a) Write the mechanism of the reaction of butanoyl chloride with propanamine, in the presence of pyridine shown below.

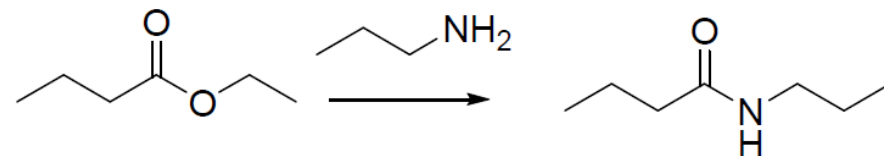


PYRIDINE COULD ALSO BE THE BASE THAT REMOVES THE PROTON IN THE SECOND STEP. THIS WAY HCl IS REMOVED EFFECTIVELY

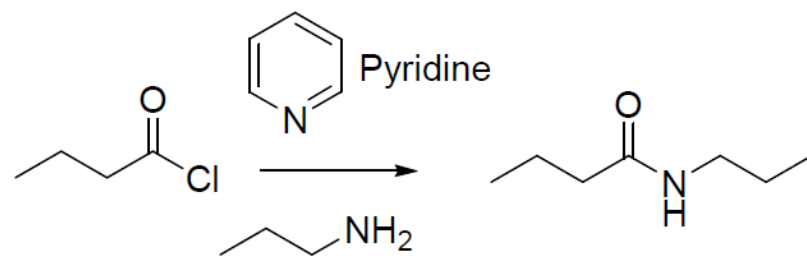
b) Explain why pyridine is used in the system (If pyridine was not used, what would be the impact on the reaction yield and why).



a) Draw the mechanism of this reaction.

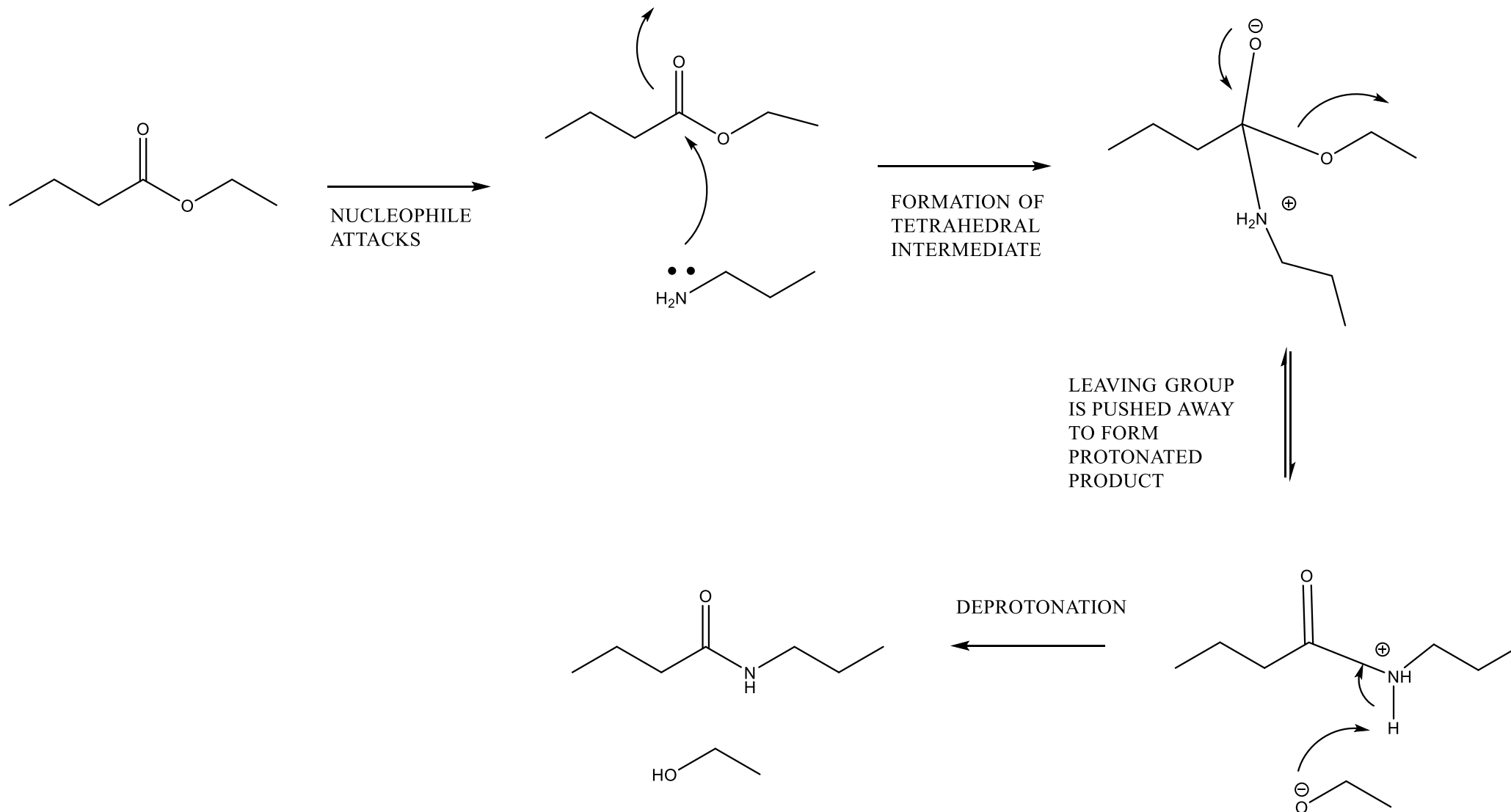
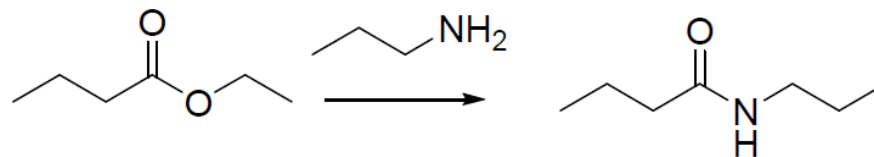


b) Do you expect this reaction to be more or less efficient kinetically and thermodynamically than the reaction below? Explain why. (Which reaction do you expect to give the best yield of the product?)

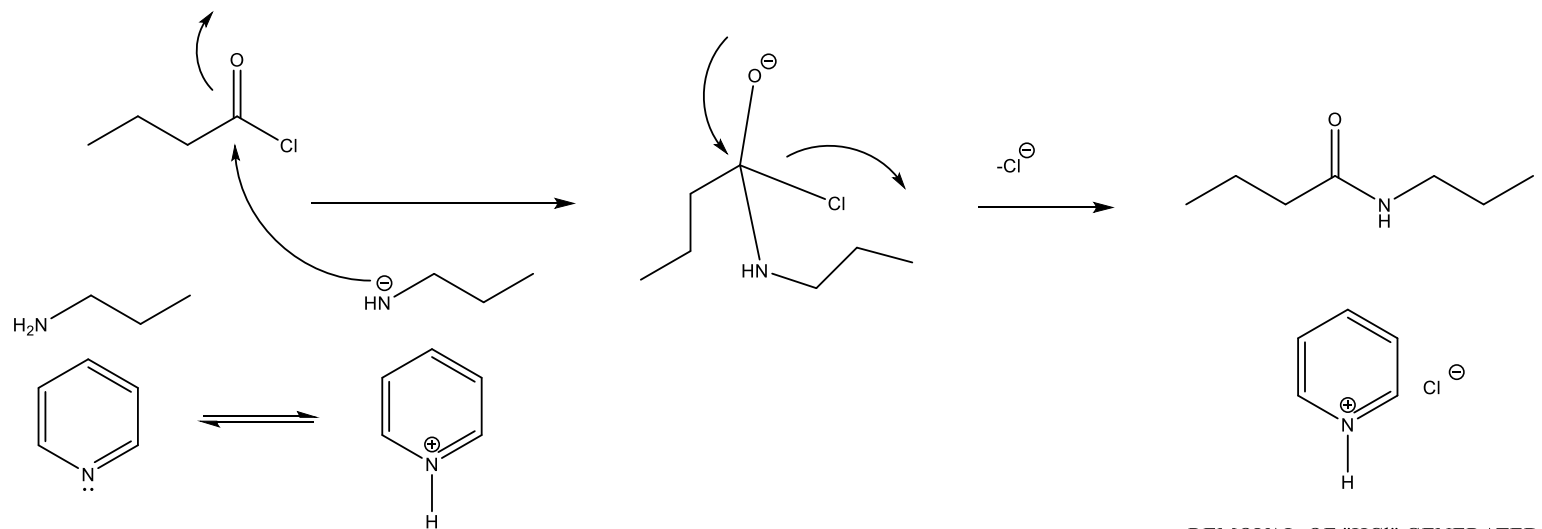
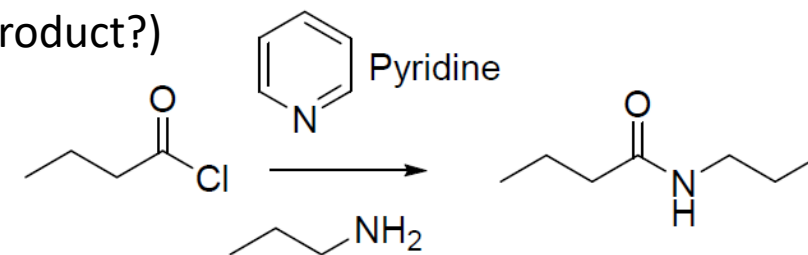


10 Minutes

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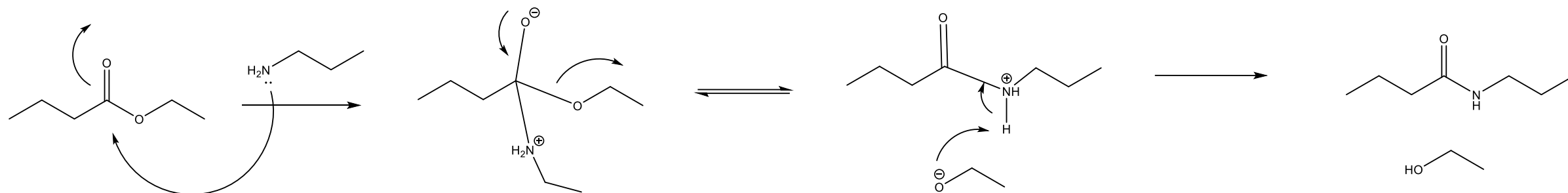


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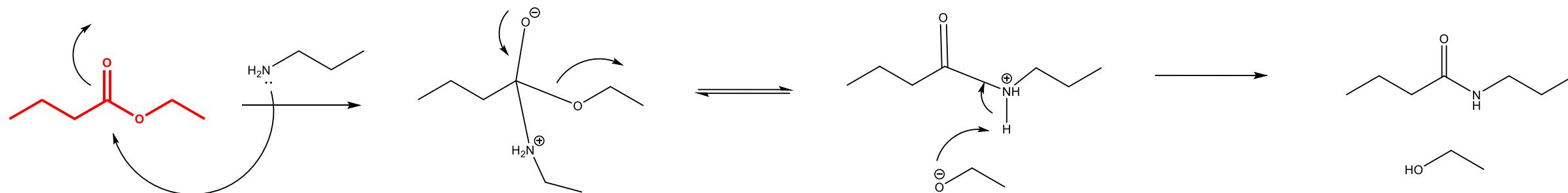
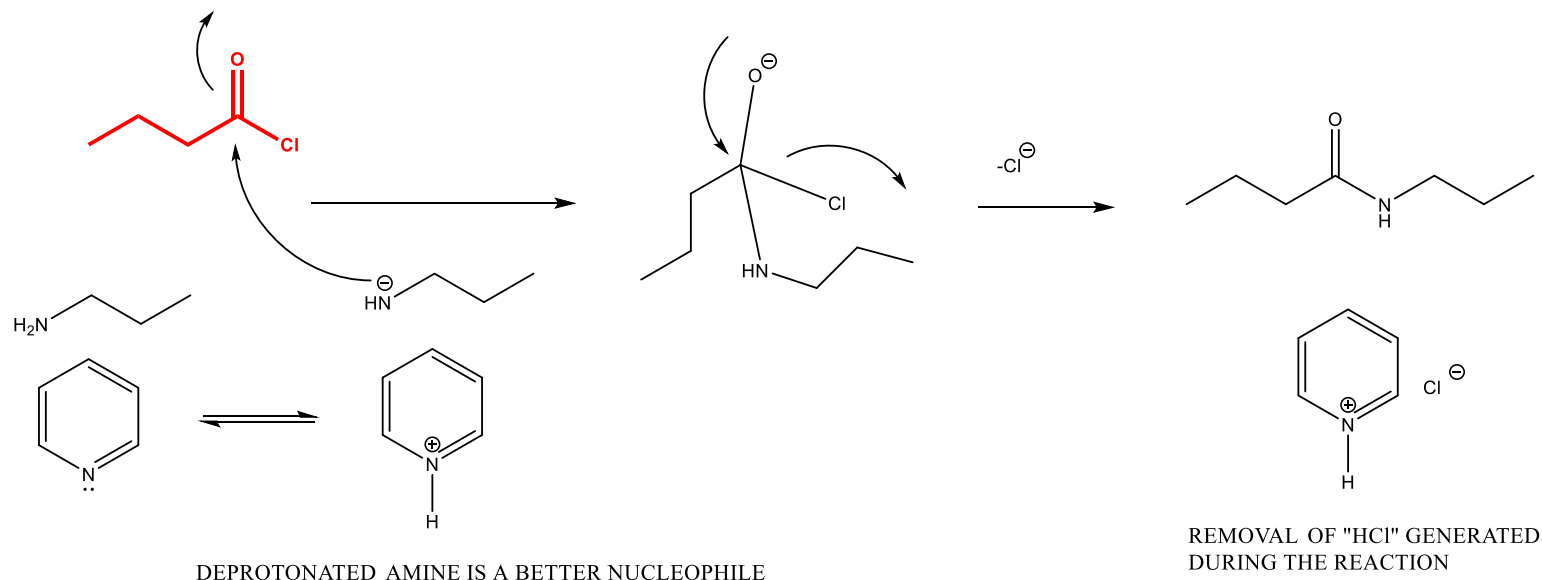
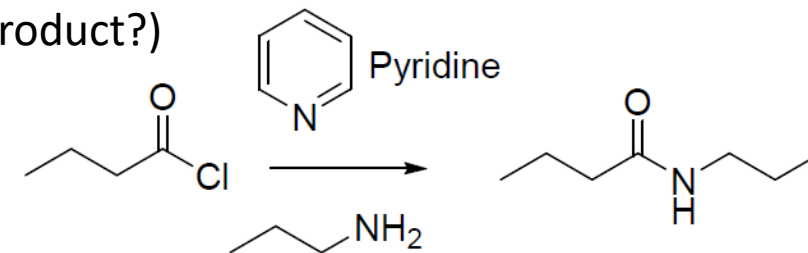


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REMOVAL OF "HCl" GENERATED DURING THE REACTION

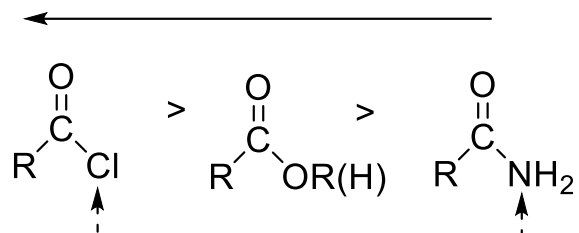


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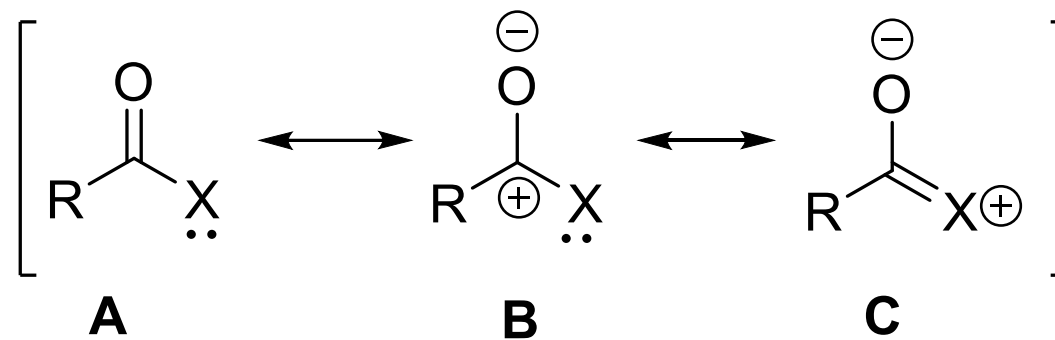
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increasing reactivity to nucleophiles



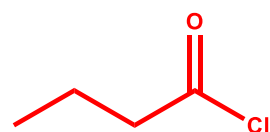
Very EN, but poor lone pair donation

EN, but good lone pair donation

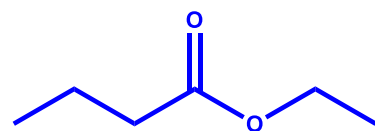


STABILITY OF CARBOXYLIC ACID DERIVATIVES
EXPLAINED IN THE LECTURE NOTES

Free Gibbs Energy

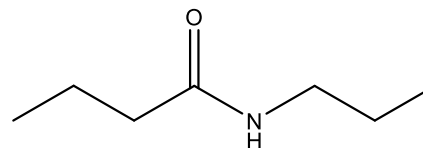


ΔG_1



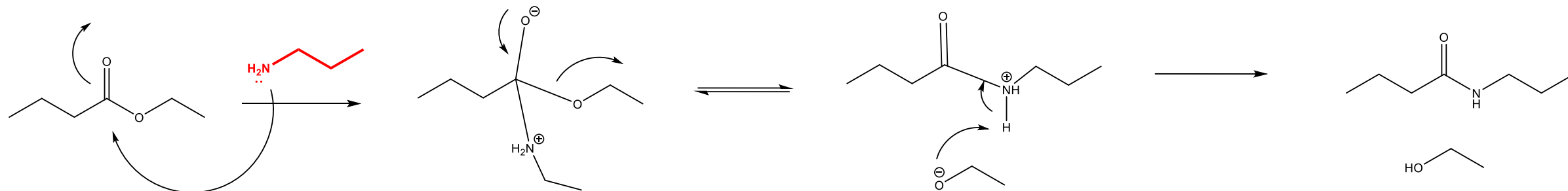
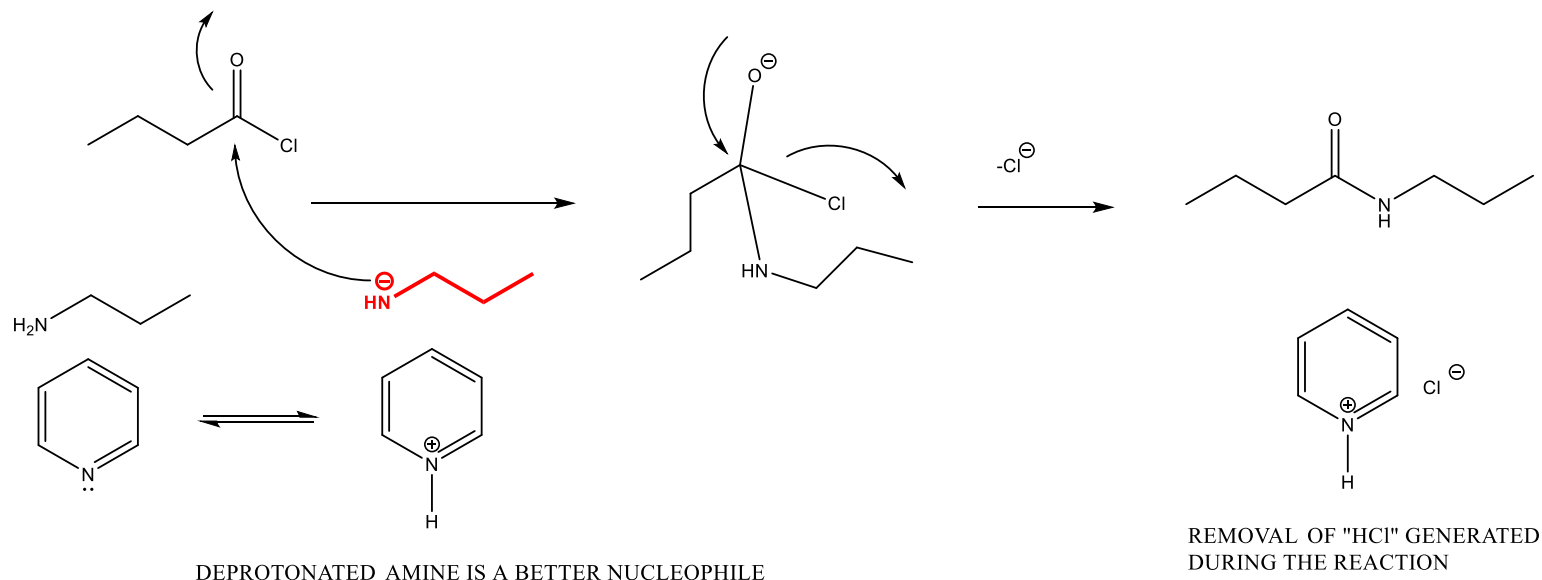
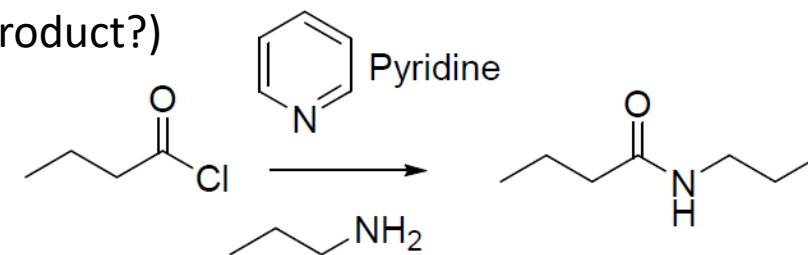
ΔG_2

$\Delta G_1 > \Delta G_2$

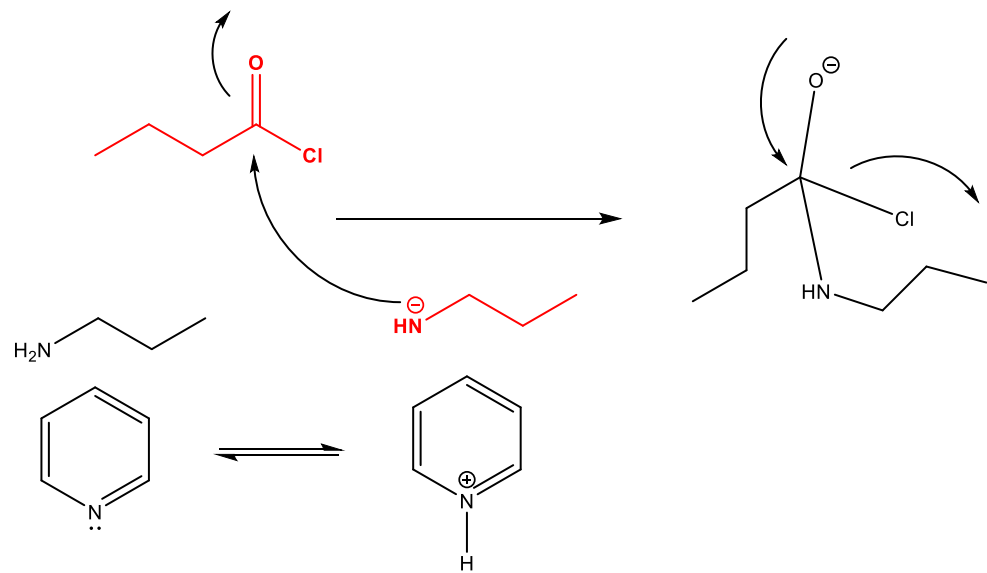


THERMODYNAMICALLY, THE REACTION VIA
ACYL CHLORIDE REACTANT IS MORE LIKELY
TO HAPPEN

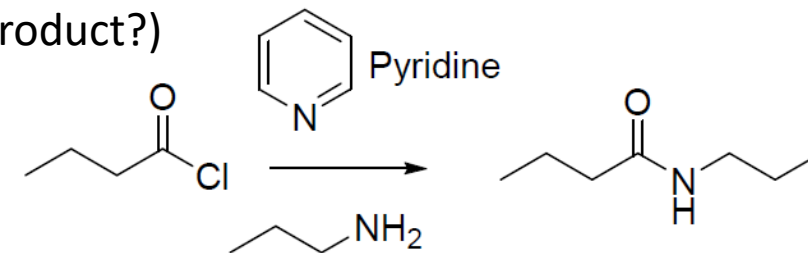
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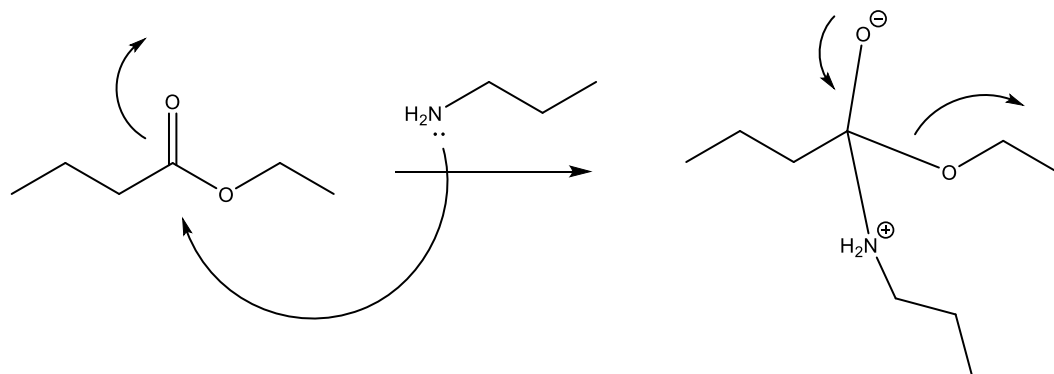
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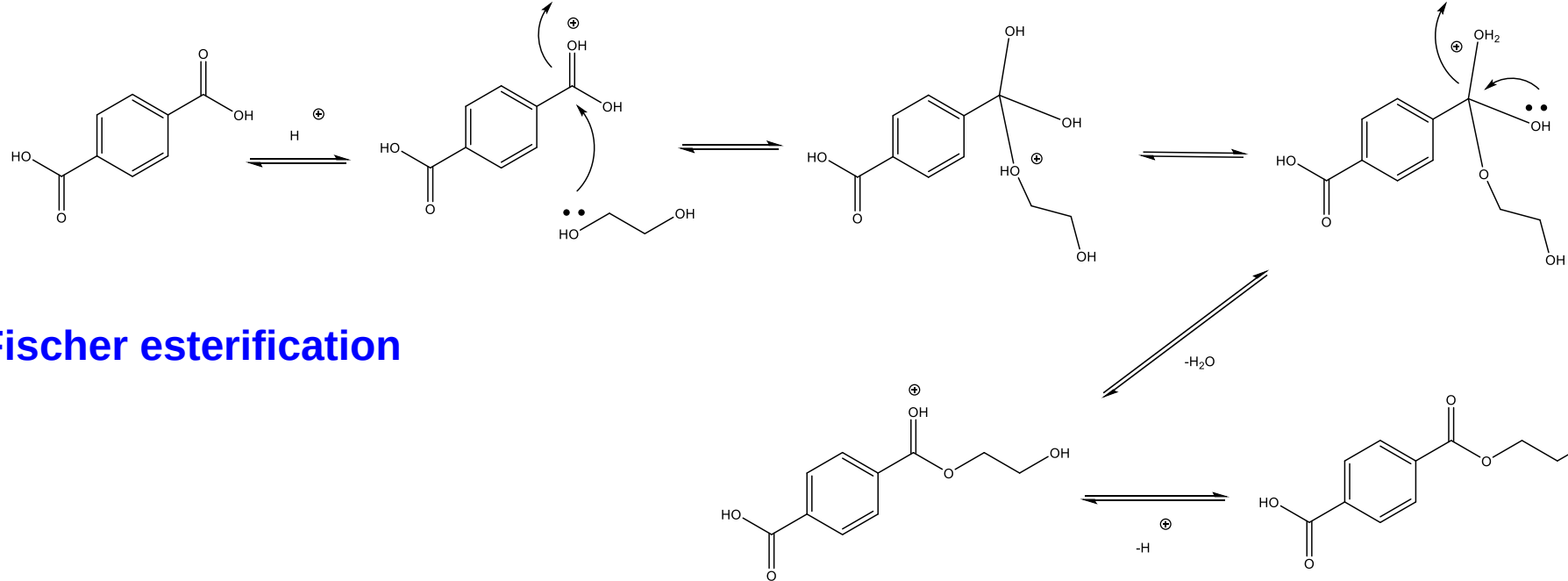


KINETICALLY, THE REACTION VIA THE MORE REACTIVE ACYL CHLORIDE REACTANT WITH THE BETTER NUCLEOPHILE (DEPROTONATED AMINE) IS MORE LIKELY TO HAPPEN. OVERALL THE REACTION STARTING WITH THE ACYL CHLORIDE IN THE PRESENCE OF PYRIDINE WILL GIVE THE BEST YIELD.

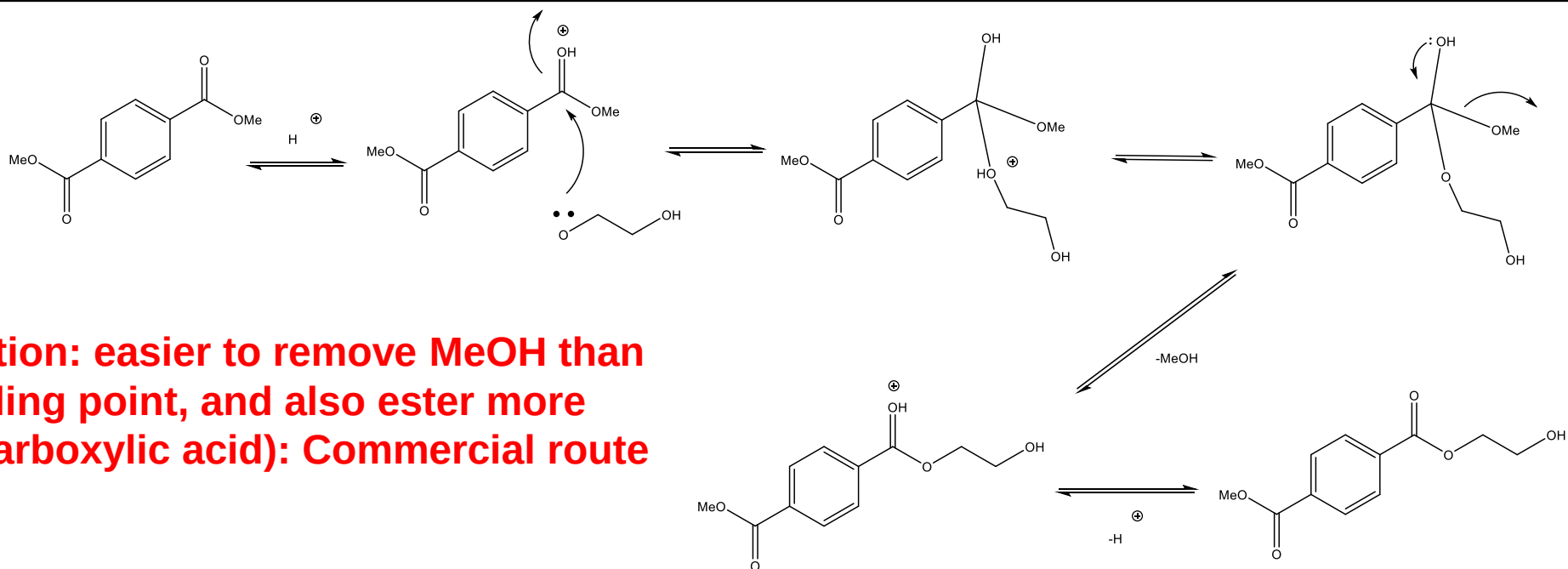


The industrial synthesis of PET is NOT the Fischer esterification, but a transesterification (see Lecture notes on esters). Explain why this is the case from a thermodynamics point of view.

10 Minutes

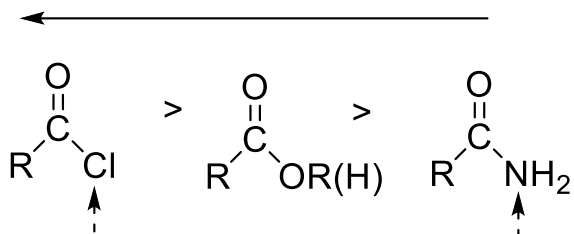


Classical Fischer esterification



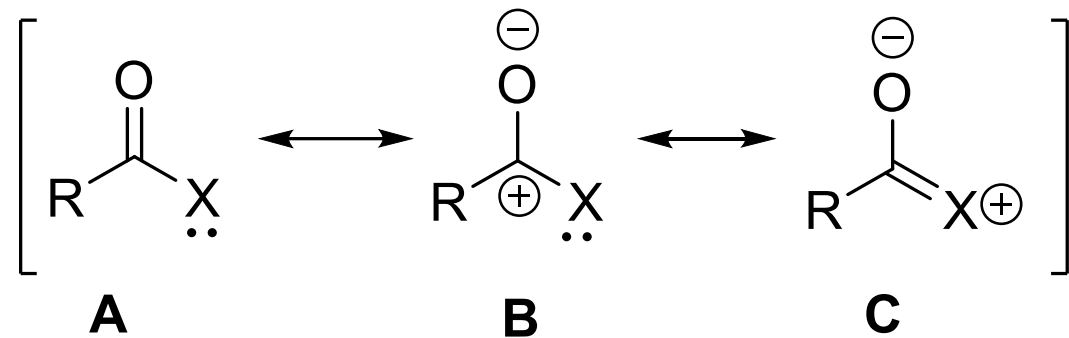
Transesterification: easier to remove MeOH than H₂O (lower boiling point, and also ester more reactive than carboxylic acid): Commercial route to PET

increasing reactivity to nucleophiles



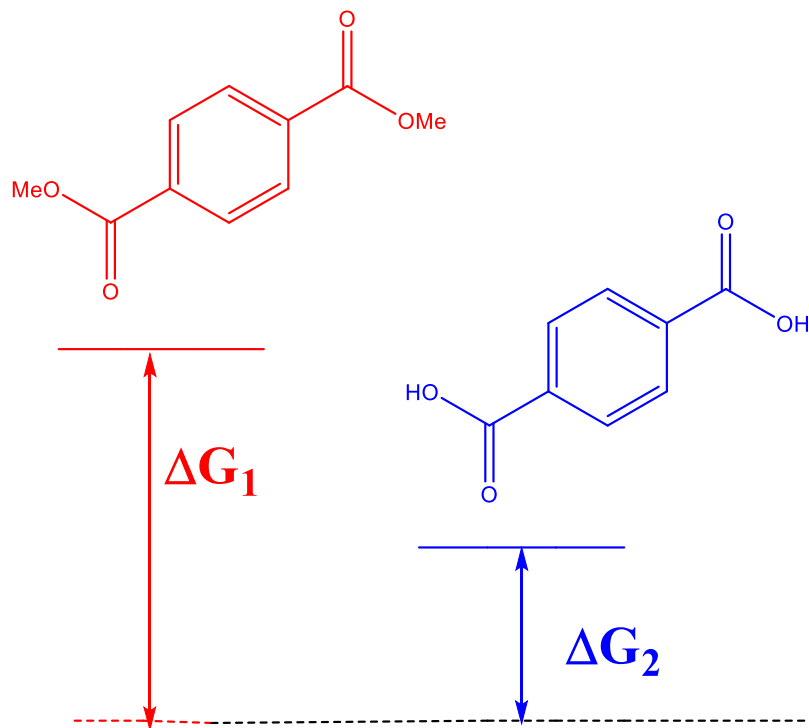
Very EN, but poor lone pair donation

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STABILITY OF CARBOXYLIC ACID DERIVATIVES
EXPLAINED IN THE LECTURE NOTES

Free Gibbs Energy



$\Delta G_1 > \Delta G_2$

THERMODYNAMICALLY, THE REACTION VIA
ESTER REACTANT IS MORE LIKELY TO
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